

Student Number:	
Class:	

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION 2013**

**MATHEMATICS
EXTENSION 2**

General Instructions:

- Reading Time: 5 minutes.
- Working Time: 3 hours.
- Write in black pen.
- Board approved calculators & templates may be used
- A Standard Integral Sheet is provided.
- In Question 11 - 16, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged working.

Total Marks 100

Section I: 10 marks

- Attempt Question 1 – 10.
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II: 90 Marks

- Attempt Question 11 - 16
- Answer on lined paper provided. Start a new page for each new question.
- Allow about 2 hours & 45 minutes for this section.

The answers to all questions are to be returned in separate *stapled* bundles clearly labelled Question 11, Question 12, etc. Each question must show your Candidate Number.

Section I

10 marks

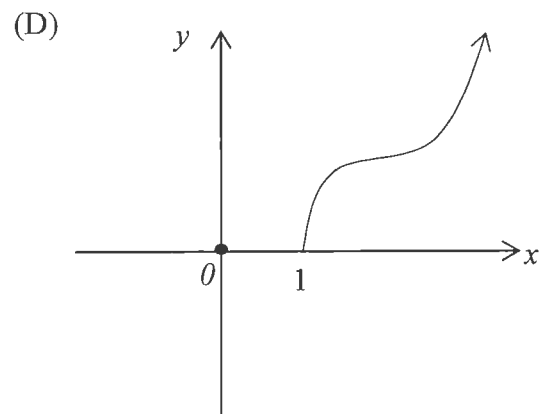
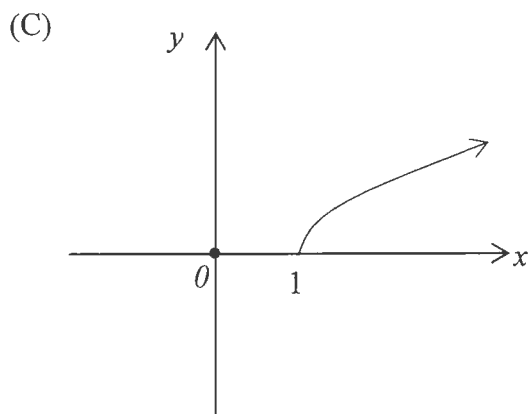
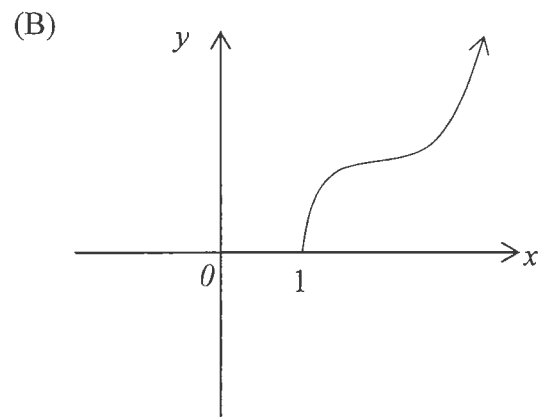
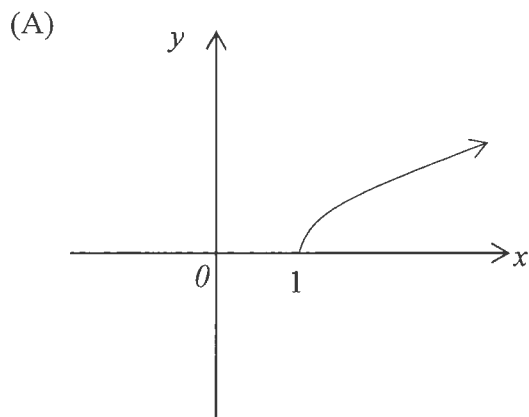
Attempt Questions 1–10

Allow about 15 minutes for this section

- 1 Let $z = 1 + i$. What is the value of z^{12} ?

- (A) 64
- (B) -64
- (C) $64i$
- (D) $-64i$

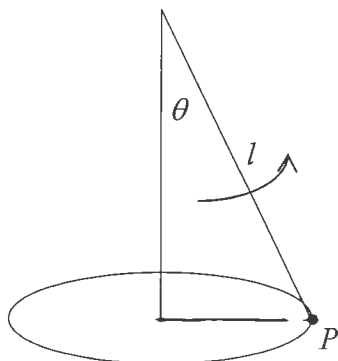
- 2 Given $f(x) = x^2(x-1)$. Which of the following best represents the graph of $y = \sqrt{f(x)}$?



- 3 Given $2x^2 + xy + 2y^2 = 30$, what are the coordinates of one of the vertical tangents?
- (A) $(-1, 4)$
- (B) $(4, -1)$
- (C) $(-1, -4)$
- (D) $(1, -4)$
- 4 What is the equation of the chord of contact of tangents from $(2, 1)$ to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$?
- (A) $\frac{2x}{9} - \frac{y}{4} = 1$
- (B) $\frac{2x}{9} + \frac{y}{4} = 1$
- (C) $\frac{x}{9} - \frac{y}{2} = 1$
- (D) $\frac{x}{9} + \frac{y}{4} = 1$
- 5 Given $3x^3 - 2x + 5 = 0$ has roots α , β and γ , what is the equation with roots $\alpha + 1$, $\beta + 1$ and $\gamma + 1$?
- (A) $3x^3 - 9x^2 + 7x + 6 = 0$
- (B) $3x^3 + 9x^2 + 7x + 6 = 0$
- (C) $3x^3 - 9x^2 + 7x + 4 = 0$
- (D) $3x^3 + 9x^2 + 7x + 4 = 0$

- 6 Which of the following is the correct expression for the integral $\int \frac{dx}{4 + \sin^2 x}$?
- (A) $\frac{1}{2\sqrt{5}} \tan^{-1} \left(\frac{5}{4} \tan x \right) + C$
- (B) $2\sqrt{5} \tan^{-1} \left(\frac{5}{4} \tan x \right) + C$
- (C) $\frac{1}{2\sqrt{5}} \tan^{-1} \left(\frac{\sqrt{5}}{2} \tan x \right) + C$
- (D) $2\sqrt{5} \tan^{-1} \left(\frac{\sqrt{5}}{2} \tan x \right) + C$
- 7 Given $3x^3 + 6x - 5 = 0$ has roots α , β and γ , what is the value of $\alpha^3 + \beta^3 + \gamma^3$?
- (A) 5
- (B) 9
- (C) 15
- (D) -1
- 8 The equation of motion of a particle falling with velocity v m/s is given by $\ddot{x} = 10 - \frac{v}{2}$. Which of the following is the value of the terminal velocity?
- (A) 5
- (B) 15
- (C) 20
- (D) $\sqrt{20}$

- 9 A bob P of mass m kg is suspended from a fixed point A by a string of length l metres, and acceleration due to gravity g . P describes a horizontal circle with uniform angular velocity ω rad/s.



Which of the following expressions represents the tension in the string?

- (A) $ml\omega$
 (B) $ml\omega^2$
 (C) $mg l \omega$
 (D) $mg l \omega^2$
- 10 Which of the following is the correct expression for the integral $\int e^{\alpha x} \sin \beta x \, dx$?

- (A) $\frac{e^{\alpha x}}{\alpha^2 + \beta^2} [\beta \sin \beta x + \alpha \cos \beta x] + C$
 (B) $\frac{e^{\alpha x}}{\alpha^2 + \beta^2} [\beta \sin \beta x - \alpha \cos \beta x] + C$
 (C) $\frac{e^{\alpha x}}{\alpha^2 + \beta^2} [\alpha \sin \beta x + \beta \cos \beta x] + C$
 (D) $\frac{e^{\alpha x}}{\alpha^2 + \beta^2} [\alpha \sin \beta x - \beta \cos \beta x] + C$

Section II

90 marks

Attempt Questions 11–16.

Allow about 2 hours and 45 minutes for this section.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW page.

(a) $|z| < 1$ and $z = \cos \theta + i \sin \theta$, where $-\pi < \theta \leq \pi$.

(i) Show $1 + z = 2 \cos \frac{\theta}{2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$. 2

(ii) z_1 and z_2 are complex numbers such that $|z_1| = |z_2| = 1$. If z_1 and z_2 have arguments α and β respectively, where $-\pi < \alpha \leq \pi$ and $-\pi < \beta \leq \pi$, show that $\frac{z_1 + z_1 z_2}{z_1 + 1}$ has

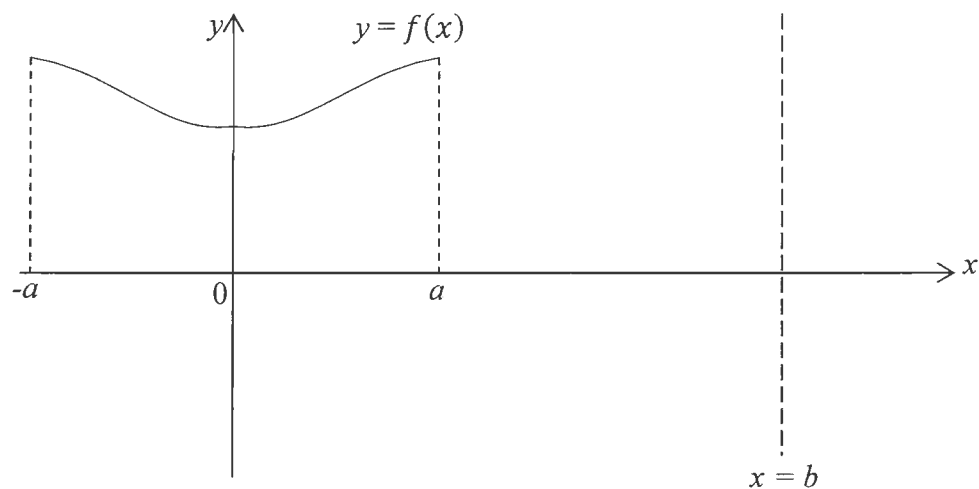
modulus $\frac{\cos \frac{\beta}{2}}{\cos \frac{\alpha}{2}}$ and Argument $\frac{\alpha + \beta}{2}$.

(iii) If $|z_1| = |z_2| = 1$ and $\frac{z_1 + z_1 z_2}{z_1 + 1} = 2i$ find z_1 and z_2 in the form $x + iy$ where x and y are real rational numbers. 4

(b) Shade the region $-\frac{\pi}{4} \leq \text{Arg } z \leq \frac{\pi}{4}$ and $|z| \leq 3$. 2

Question 11 (c) is continued over the page.

(c)

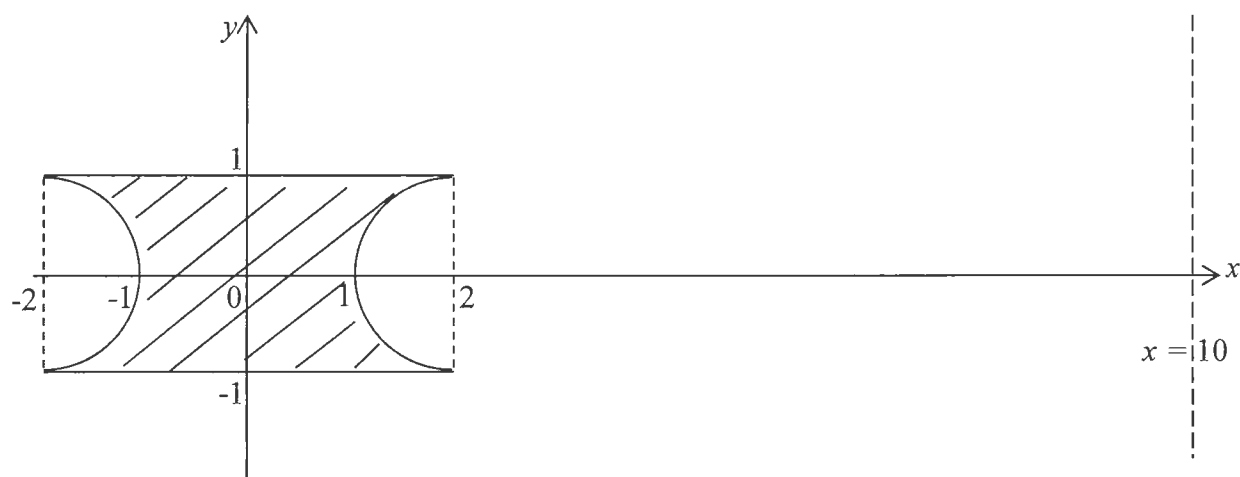


$f(x)$ is an even function such that $f(x) \geq 0$ for $-a \leq x \leq a$.

The region bounded by $y = f(x)$, the x -axis, and the ordinates $x = -a$ and $x = a$ has area A . The region is rotated about the line $x = b$ where $b > a > 0$.

(i) Using the method of cylindrical shells show that the volume V of rotation is $2\pi bA$. 3

(ii)



The region shown with circular ends is rotated about $x = 10$ to form a circular sealing ring. Find the volume of revolution. 2

End of Question 11.

Question 12 (15 marks) Start a NEW page.

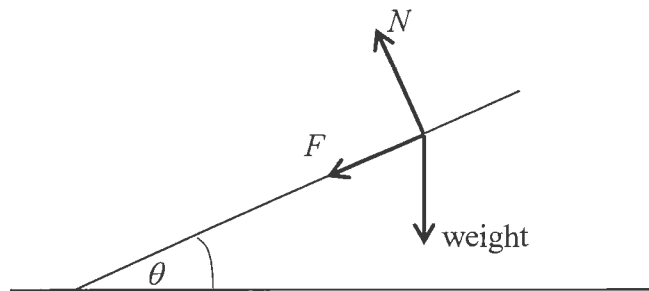
- (a) Graph $y = \frac{x}{(x+4)(x+2)}$ showing all intercepts with the coordinate axes and all asymptotes. 3

- (b) The region bounded by $y = \frac{x}{(x+4)(x+2)}$, the x -axis and $x = 1$ is rotated around the y -axis.

- (i) Find the values A , B and C such that $\frac{x^2}{(x+4)(x+2)} = A + \frac{B}{x+2} + \frac{C}{x+4}$. 4

- (ii) Using the method of cylindrical shells show that the volume V of revolution is given by $V = 2\pi \int_0^1 \frac{x^2 dx}{(x+4)(x+2)}$, hence find the exact value of the volume of revolution. 4

- (c)



- A car of mass 2000 kg travels around a curve of radius 150 m at a speed of 110km/h. The car experiences a lateral resistance force F of $0.22 \times$ normal force, N , as shown. 4

By resolving the forces vertically and horizontally find the ~~minimum~~ angle θ ~~(to the nearest minute)~~ for the car to negotiate the curve. (Assume acceleration due to gravity of 10 m/s^2).

End of Question 12.

Question 13 (15 marks) Start a NEW page.

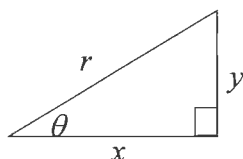
(a) (i) Show $\int_{-a}^0 f(x)dx = \int_0^a f(-x)dx$ 1

(ii) Deduce $\int_{-a}^a f(x)dx = \int_0^a [f(x) + f(-x)] dx$ 1

(iii) Hence evaluate $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{(1 + \sin x)^2}$ 4

(b) A shape is defined as $r = \frac{9}{5 + 4 \cos \theta}$ where r is the distance from origin and θ is the angle anticlockwise from the positive x -axis.

(i) Using the notation 3



find the equivalent Cartesian equation and show that the shape is an ellipse translated.

(ii) State the minor axis, major axis and location of the foci. 4

(iii) The area A enclosed by the shape is given by $A = \int_0^{2\pi} \frac{1}{2} r^2 d\theta$. 2

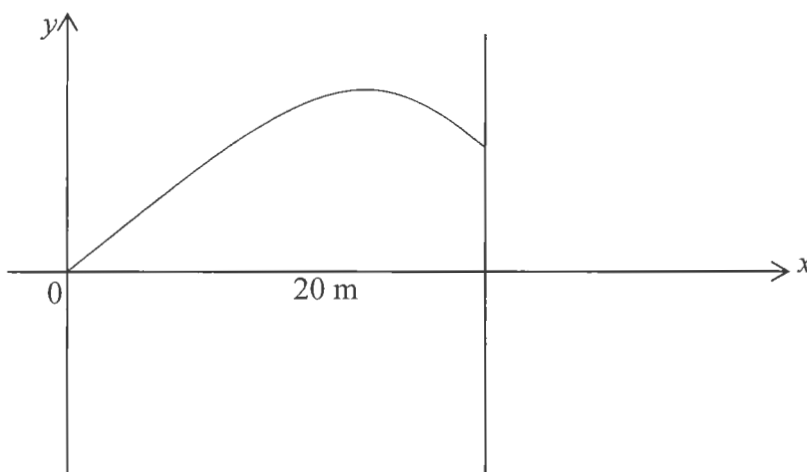
Using (b)(i) and (b)(ii) evaluate $\int_0^{2\pi} \frac{d\theta}{(5 + 4 \cos \theta)^2}$.

End of Question 13.

Question 14 (15 marks) Start a NEW page.

- (a) (i) Find the coordinates of the intersection of the curves $y^2 = 8x$ and $x^2 = 8y$. 1
- (ii) The base of a solid is in the region bounded by the curves $y^2 = 8x$ and $x^2 = 8y$, and its cross sections by planes perpendicular to the x -axis are semicircles. Find the volume of the solid. 3

(b)



A liquid particle of mass m kg is projected from the ground and hits a vertical wall 20m from the point of projection as shown.

- (i) The equations of motion before the particle hits the wall are 3
- $$x = 4t \text{ and } y = 30t - 5t^2$$
- where t is time in seconds. Show that the particle hits the wall 25 m above the ground with a downwards velocity of 20 m/s.
- (ii) After hitting the wall the particle slides down the wall with a resistance force equal to $0.04mv^2$.
- (α) If acceleration due to gravity is 10 m/s^2 show that the velocity on return to the ground is approximately 16.44 m/s. 4
- (β) Find the total time for the particle to return to the ground. Give your answer to two decimal places. 4

End of Question 14.

Question 15 (15 marks) Start a NEW page.

The hyperbola $xy = c^2$ meets the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $P\left(ct_1, \frac{c}{t_1}\right)$ and $Q\left(ct_2, \frac{c}{t_2}\right)$ where $t_1 > t_2 > 0$. Tangents to the hyperbola at P and Q meet at T , while tangents to the ellipse at P and Q meet at V .

(i) Show the above information on a sketch. 1

(ii) Show that the parameter of point $\left(ct, \frac{c}{t}\right)$ which lies on the intersection of 2

$$xy = c^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ satisfies the equation } b^2c^2t^4 - a^2b^2t^2 + a^2c^2 = 0.$$

(iii) Given the equation of the tangent to the hyperbola at $\left(ct, \frac{c}{t}\right)$ is $x + t^2y = 2ct$, show 2

$$\text{that the coordinates of } T \text{ are } \left(\frac{2ct_1t_2}{t_1+t_2}, \frac{2c}{t_1+t_2}\right).$$

(iv) Given that the equation of the tangent to the ellipse at (x_1, y_1) is $b^2x_1x + a^2y_1y = a^2b^2$, 2

$$\text{show that the coordinates of } V \text{ are } \left(\frac{a^2}{c(t_1+t_2)}, \frac{b^2t_1t_2}{c(t_1+t_2)}\right).$$

(v) Show that the line TV passes through the origin. 3

(vi) Point V lies at a focus of the hyperbola.

(α) Show that the ellipse is a circle. 2

(β) Find the radius of the circle in terms of c . 3

End of Question 15.

Question 16 (15 marks) Start a NEW page.

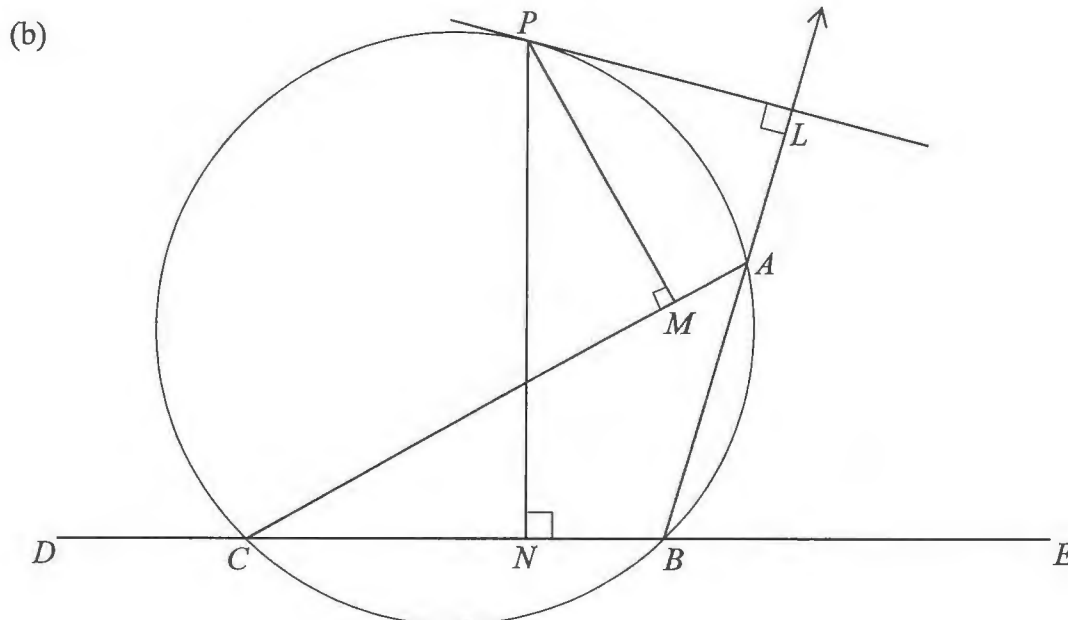
(a) $I_n = \int_0^{2\pi} (1 + \cos \theta)^n d\theta$ for $n \geq 0$.

(i) Show $I_{n+1} = \frac{2n+1}{n+1} I_n$.

4

(ii) Find I_3 .

1



ABC is a triangle inscribed in a circle. L , M and N are the feet of the perpendiculars from P to AB , AC and BC respectively.

(i) Copy the diagram.

1

(ii) Show P , M , A and L are concyclic points.

2

(iii) Show P , C , N and M are concyclic points.

2

(iv) Show that L , M and N are collinear.

5

End of paper.

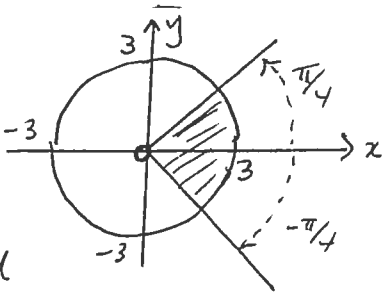
» **Section I**

1 mk for each question.

1. A
2. D
3. B
4. A
5. C
6. C
7. A
8. C
9. B
10. D

Suggested Solutions	Marks	Marker's Comments
(a) $1+z = (1+\cos\theta) + i(\sin\theta)$ (i) $= (1+\cos 2 \times \frac{\theta}{2}) + i(\sin 2 \times \frac{\theta}{2})$ $= 2\cos^2 \frac{\theta}{2} + i(2\sin \frac{\theta}{2} \cos \frac{\theta}{2})$ $= 2\cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})$ or $2\cos \frac{\theta}{2} \operatorname{cis} \frac{\theta}{2}$	1 1	This part was well done by most students
(ii) $z_1 = 2\cos \frac{\alpha}{2} \operatorname{cis} \frac{\alpha}{2}$ $z_2 = 2\cos \frac{\beta}{2} \operatorname{cis} \frac{\beta}{2}$ $\left \frac{z_1(1+z_2)}{1+z_1} \right = \frac{ z_1 1+z_2 }{ 1+z_1 }$ $= \frac{(1)(2\cos \frac{\beta}{2})}{(2\cos \frac{\alpha}{2})}$ $= \frac{\cos \frac{\beta}{2}}{\cos \frac{\alpha}{2}}$ NOTE This is positive since $-\pi < \alpha, \beta < \pi$ $\arg \left(\frac{z_1(1+z_2)}{1+z_1} \right) = \arg z_1 + \arg(1+z_2) - \arg(1+z_1)$ $= \alpha + \frac{\beta}{2} - \frac{\alpha}{2}$ $= \frac{\alpha + \beta}{2}$	1 1	Quite a few failed to factorise $z_1 + z_2, z_2$ and hence did not use a(i) which made the question more difficult. A significant number of students confused $\cos \frac{\alpha}{2}$ and $\operatorname{cis} \frac{\alpha}{2}$

MATHEMATICS Extension 2: Question. 1

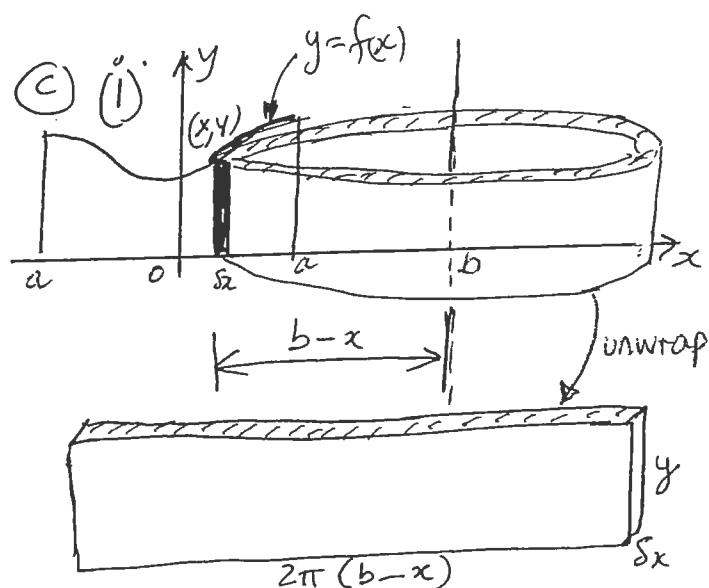
Suggested Solutions	Marks	Marker's Comments
(iii) $\arg(2i) = \frac{\pi}{2}$ $\therefore \frac{\alpha + \beta}{2} = \frac{\pi}{2}$ $\alpha + \beta = \pi$ $2i = 2$ $\frac{\cos \beta/2}{\cos \alpha/2} = 2$ $\frac{\cos(\frac{\pi}{2} - \alpha/2)}{\cos \alpha/2} = \frac{\sin \frac{\alpha}{2}}{\cos \alpha/2} = 2$ $t = \tan \frac{\alpha}{2} = 2$ $\cos \alpha = \frac{1-t^2}{1+t^2} = \frac{1-2^2}{1+2^2} = -\frac{3}{5}$ $\sin \alpha = \frac{2t}{1+t^2} = \frac{2 \times 2}{1+2^2} = \frac{4}{5}$ $\therefore z_1 = -\frac{3}{5} + \frac{4}{5}i$ Similarly $\tan \beta/2 = \frac{1}{2}$ $\cos \beta = \frac{3}{5}$ $\sin \beta = \frac{4}{5}$ $z_2 = \frac{3}{5} + \frac{4}{5}i$	1 1 1 1	Some thought that the $\arg 2i = \pi$ Many missed the fact that $\cos(\frac{\pi}{2} - \frac{\alpha}{2}) = \sin \frac{\alpha}{2}$ ← This mark for $\tan \frac{\alpha}{2} = 2$ or $\tan \beta/2 = \frac{1}{2}$ Many made arithmetic mistakes or assumed things like $z_1 = -z_2$ or $z_1 = z_2$ Full marks for z_1 & z_2 correctly obtained.
(b)  NOTE $(0,0)$ excluded	1 1	Some students did not note the $\frac{\pi}{4}$ angles. Quite a few did not note that the origin is excluded

MATHEMATICS Extension 2: Question. //

Suggested Solutions

Marks

Marker's Comments



$$\text{Volume} = \lim_{\delta x \rightarrow 0} 2\pi(b-x)y \delta x$$

$$V = \int_{-a}^a 2\pi f(x)(b-x) dx \quad *$$

$$V = 2\pi b \int_{-a}^a f(x) dx - 2\pi \int_{-a}^a x f(x) dx$$

$$V = 2\pi b A \quad \text{since } \int_{-a}^a f(x) dx = A$$

$$\text{and } \int_{-a}^a x f(x) dx = 0 \quad \text{since } x f(x) \text{ is odd} \\ (\text{odd} \times \text{even} = \text{odd}) \\ (\text{func.} \quad \text{func.} \quad \text{func.})$$

$$(ii) \text{ From (i) } V = 2\pi b A$$

$$A = \text{rectangle} - \text{circle} \\ = 4 \times 2 - \pi(1)^2 \\ = 8 - \pi$$

$$b = 10$$

$$V = 2\pi(10)(8 - \pi)$$

$$V = 20\pi(8 - \pi) \text{ units}^3$$

Some students stated * without justification and were not awarded full marks

Students needed to explain why $\int_{-a}^a f(x) dx = A$ and $\int_{-a}^a x f(x) dx = 0$

Many students wasted time by not using (i) but by finding the volume by integration.

A common error was to think that the area of the rectangle was 4

Suggested Solutions

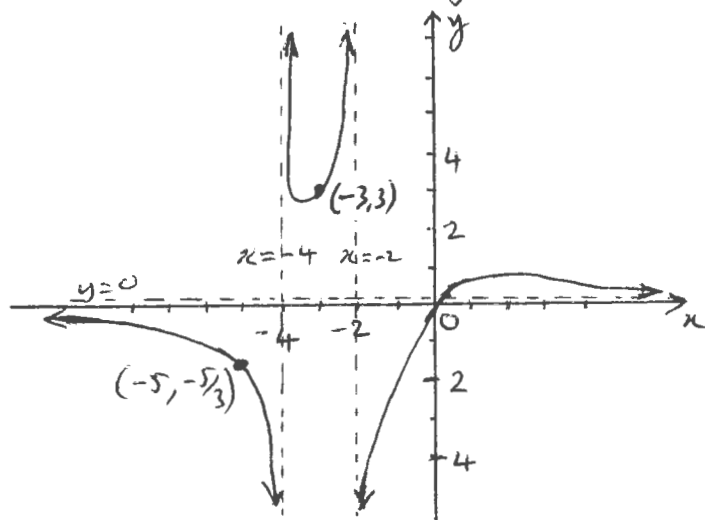
Marks

Marker's Comments

a) Vert Asy $x = -4, x = -2$

Hor. Asy $y = 0$

Zero at $x = 0$, also y intercept.



3x 1/2

for each asymptote with either an equation or a line definitely finishing towards it.

1

for shape - 1/2 off for each real error in main graph.

1/2

for having a labelled point on each branch, (or associated scales)

$$b) i) \frac{x^2}{(x+4)(x+2)} = A + \frac{B}{x+2} + \frac{C}{x+4}$$

$$\therefore A(x+2)(x+4) + B(x+4) + C(x+2) \equiv x^2$$

$$\text{Equate coeffs. of } x^2 : \underline{\underline{A = 1}}$$

$$\text{Put } x = -2 : 2B = 4 \therefore \underline{\underline{B = 2}}$$

$$\text{Put } x = -4 : -2C = 16 \therefore \underline{\underline{C = -8}}$$

1

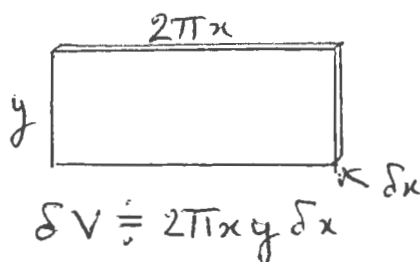
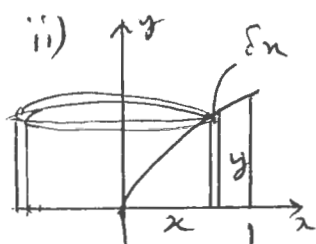
Easy marks.

1

1

1

ii)



$$\delta V \doteq 2\pi x y \delta x$$

$$\delta V = (\pi(x+\delta x)^2 - \pi x^2) y$$

$$= 2\pi x y \delta x \text{ (neglecting 2nd order terms)}$$

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi x y \delta x$$

$$= 2\pi \int_0^1 x^2 dx$$

1/2

diagram

1/2

for $\doteq$ (type 2) or neglect 2nd order terms

1/2

for limit of sum

1/2

for integral (except if boldly stated)

Suggested Solutions

Marks

Marker's Comments

b) ii) (cont)

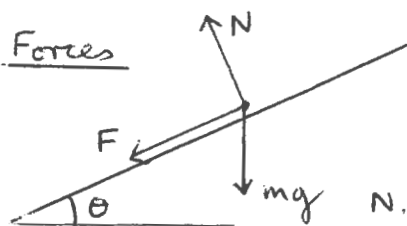
$$\therefore V\Omega = 2\pi \int_0^1 \left(1 + \frac{2}{x+2} - \frac{8}{x+4}\right) dx$$

(using part i)

$$= 2\pi \left[x + 2\ln(x+2) - 8\ln(x+4) \right]_0^1$$

$$= 2\pi \{1 + 2\ln 3 - 8\ln 5 - 2\ln 2 + 8\ln 4\}$$

$$\underline{V\Omega = 2\pi (1 + 2\ln 3 + 14\ln 2 - 8\ln 5) \text{ m}^3}$$

c) ForcesAccⁿ

$$\leftarrow \frac{v^2}{r}$$

$$\text{N.B. } v = 110 \text{ k/h} = \frac{275}{9} \text{ m/s}$$

Resolve vertically (V) $mg + F\sin\theta = N\cos\theta$

Resolve horizontally (H) $F\cos\theta + N\sin\theta = \frac{mv^2}{r}$

(Assuming $F = 0.22 \text{ N}$ means this is already the optimal angle θ .)

Substituting numbers

$$(V) \rightarrow N(\cos\theta - 0.22\sin\theta) = 20000$$

$$(H) \rightarrow N(0.22\cos\theta + \sin\theta) = 12448.56$$

Dividing:

$$\frac{\cos\theta - 0.22\sin\theta}{0.22\cos\theta + \sin\theta} = 1.6066\dots$$

$$\frac{1 - 0.22\tan\theta}{0.22 + \tan\theta} = 1.6066\dots$$

$$\tan\theta = 0.3539$$

$$\underline{\underline{\theta = 19^\circ 29' \text{ (nearest minute)}}}$$

1/2

Most people got these 2 1/2 marks.

1

1

Many mistakes in the numeric work. 1/2 for getting down to a simple equation in $\tan\theta$.

1/2

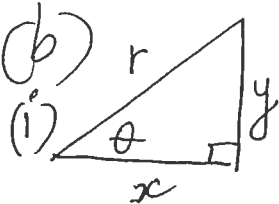
Final mark for correct solution (29' or 30' accepted)

1

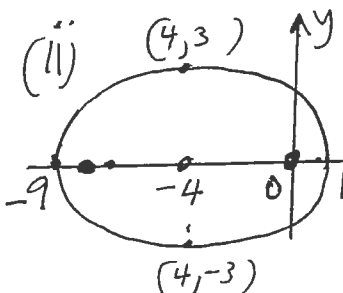
MATHEMATICS Extension 2: Question.

Suggested Solutions	Marks	Marker's Comments
(a) (i) Let $\left. \begin{array}{l} x = -u \\ dx = -du \\ x=0 \quad u=0 \\ x=-a \quad u=a \end{array} \right\}$ $\begin{aligned} \therefore \int_{-a}^0 f(x) dx &= \int_a^0 f(-u) (-du) \\ &= \int_0^a f(-u) du \\ &= \int_0^a f(x) dx \end{aligned}$ Changing the variable in a definite integral does not change its value	1	Well done by students Some students thought that the function must be even (or must be odd)
(ii) $\int_{-a}^a f(x) = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$ $\begin{aligned} &= \int_0^a f(-x) dx + \int_0^a f(x) dx \\ &= \int_0^a [f(x) + f(-x)] dx \end{aligned}$	1	Well done by students
(iii) $\int_{-\pi/4}^{\pi/4} \frac{dx}{(1+\sin x)^2} = \int_0^{\pi/4} \frac{1}{(1+\sin x)^2} + \frac{1}{[1+\sin(-x)]^2} dx$ $\begin{aligned} &= \int_0^{\pi/4} \frac{1}{(1+\sin x)^2} + \frac{1}{(1-\sin x)^2} dx \\ &= \int_0^{\pi/4} \frac{(1-\sin x)^2 + (1+\sin x)^2}{(1-\sin^2 x)^2} dx \end{aligned}$	1	Nearly all students used a(ii) correctly to begin

MATHEMATICS Extension 2: Question.

Suggested Solutions	Marks	Marker's Comments
$\int_{-\pi/4}^{\pi/4} \frac{dx}{(1+\sin x)^2} = \int_0^{\pi/4} \frac{2(1+\sin^2 x)}{\cos^4 x} dx \quad (A)$ $= 2 \int_0^{\pi/4} \sec^2 x (\sec^2 x + \tan^2 x) dx$ $= 2 \int_0^{\pi/4} \sec^2 x (1 + 2\tan^2 x) dx$ $= 2 \left[\tan x + \frac{2}{3} \tan^3 x \right]_0^{\pi/4}$ $= 10/3$	1 1 1	Most students got to (A). Many failed to realise $\int \sec^2 x \tan^2 x dx = \frac{1}{3} \tan^3 x$ Correct answer correctly done (by many of a variety of methods) for full marks
(b)  (i) $r = \sqrt{x^2 + y^2}$ $\cos \theta = \frac{x}{r}$ $r = \frac{9}{5 + 4 \cos \theta}$ $r = \frac{9}{5 + 4(\frac{x}{r})}$ $1 = \frac{9}{5r + 4x}$ $5r = 9 - 4x$ $25r^2 = 81 - 72x + 16x^2$ $25(x^2 + y^2) = 81 - 72x + 16x^2$ $9x^2 + 72x + 25y^2 = 81$ $9(x^2 + 8x + 16) + 25y^2 = 81 + 9 \times 16$ $9(x+4)^2 + 25y^2 = 225$ $\frac{(x+4)^2}{25} + \frac{y^2}{9} = 1$ This is an ellipse, centre $(-4, 0)$	1 1 1	Most students failed to eliminate both θ and r so we unable to make progress Arithmetic mistakes were common here Complete simplification required for full marks.

MATHEMATICS Extension 2: Question..

Suggested Solutions	Marks	Marker's Comments
(II)  MAJOR AXIS = $2 \times 5 = 10$ units MINOR AXIS = $2 \times 3 = 6$ units $e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{4}{5}$ $ae = 4 \times \frac{4}{5} = 4$ FOCI: $(-4 \pm 4, 0) \Rightarrow (-8, 0)$ and $(0, 0)$	1 1 1 1	Many students confused semiaxis (a or b) with axis (2a or 2b)
(III) AREA = $\int_0^{2\pi} \frac{1}{2} r^2 d\theta$ $\pi ab = \int_0^{2\pi} \frac{1}{2} \left(\frac{9}{5+4\cos\theta} \right)^2 d\theta$ Area = $\pi \times 3 \times 5 = 15\pi$ $\therefore \int_0^{2\pi} \frac{d\theta}{(5+4\cos\theta)^2}$ $= 15\pi \times \frac{2}{81}$ $= \frac{10\pi}{27}$	1 1	Many wasted time finding the area of the ellipse by integration instead of quoting $A = \pi ab$ Full marks for correct answer correctly obtained.

MATHEMATICS Extension 2 : Question ... 14

Suggested Solutions

Marks

Marker's Comments

14 a) (i) Find the points of intersection of $x^2 = 8y$ and $y^2 = 8x$

$$x^4 = (8y)^2$$

$$x^4 = 64 \times 8x$$

$$x^4 = 512x$$

$$x^4 - 512x = 0$$

$$x(x^3 - 512) = 0$$

$$x(x-8)(x^2 + 8x + 64) = 0$$

$$x = 0$$

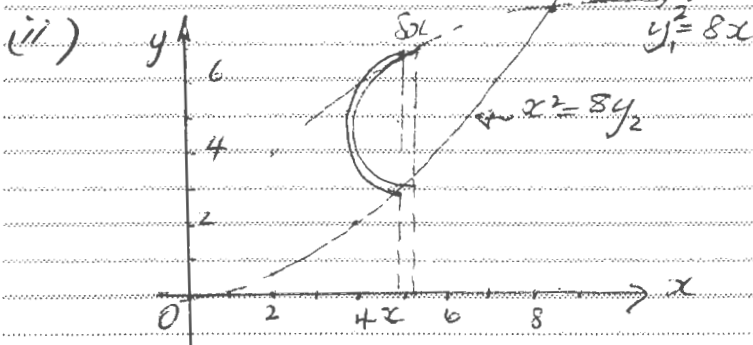
$$\text{or } x = 8$$

or resolution

$$\therefore y = 0$$

$$y = 8$$

$$P(8,8)$$



$$\begin{aligned} \text{Area of Cross Section} &= \frac{1}{2} (\pi D^2) \\ &= \frac{\pi}{8} (y_1 - y_2)^2 \\ &= \frac{\pi}{8} (2\sqrt{2x} - \frac{x^2}{8})^2 \end{aligned}$$

$$\text{Volume of Slice} = A \cdot \delta x$$

$$\text{Volume} = \lim_{\delta x \rightarrow 0} \sum_0^8 A(x) \cdot \delta x$$

$$\begin{aligned} \therefore V &= \frac{\pi}{8} \int_0^8 (2\sqrt{2x} - \frac{x^2}{8})^2 dx \\ &= \frac{\pi}{8} \int_0^8 8x - \frac{\sqrt{2}x^{5/2}}{2} + \frac{x^4}{64} dx \\ &= \frac{\pi}{8} \left[4x^2 - \frac{\sqrt{2}x^{7/2}}{7} + \frac{x^5}{320} \right]_0^8 \\ &= \frac{\pi}{8} \left[4(8)^2 - \frac{\sqrt{2}(8)^{7/2}}{7} + \frac{8^5}{320} - 0 \right] \\ &= 32\pi \left[1 - \frac{8}{7} + \frac{2}{5} \right] \\ \therefore V &= \frac{288\pi}{35} \text{ units}^3 \end{aligned}$$

MATHEMATICS Extension 2 : Question...14.

Suggested Solutions

Marks

Marker's Comments

14 b) Given $x = 4t$ and $y = 30t - 5t^2$
the particle hits the wall when
 $x = 20\text{m}$ and $y = 25\text{m}$

$$(i) \quad 4t = 20$$

$$t = 5$$

$$y = 30 \times 5 - 5(5)^2$$

$$= 150 - 125$$

$$= 25$$

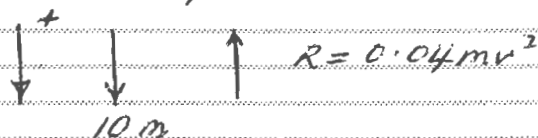
3

$\therefore$ The particle does hit the wall
25m above the ground.

And, $y' = 30 - 10t$
 $= 30 - 10 \times 5$
 $= -20$

$\therefore$ The particle has a downwards
velocity of 20 m/s

(ii). After hitting the wall, the particle
slides down with a resistance
force of $0.04mv^2$



Sum of the forces $= m\ddot{x}$ (Newton's 2nd Law)

$$\text{ie } m\ddot{x} = 10m - 0.04mv^2$$

$$\ddot{x} = 10 - 0.04v^2$$

$$\therefore \ddot{x} = -0.04(v^2 - 250) \quad v > \sqrt{250}$$

(x) For velocity on return to the ground

$$v \cdot \frac{dv}{dx} = -0.04(v^2 - 250)$$

$$\int_{20}^{25} \frac{v \cdot dv}{v^2 - 250} = \int_0^{25} -0.04 \cdot dx$$

$$\left[\frac{1}{2} \ln(v^2 - 250) \right]_{20}^{25} = -0.04 \left[x \right]_0^{25}$$

$$\frac{1}{2} [\ln(v^2 - 250) - \ln(20^2 - 250)] = -0.04 [25 - 0]$$

$$\frac{1}{2} \ln \left(\frac{v^2 - 250}{150} \right) = -1$$

4

MATHEMATICS Extension 2 : Question 14.

Suggested Solutions

Marks

Marker's Comments

14 (b)(ii) (a) continued...

$$\ln\left(\frac{v^2-250}{150}\right) = -2$$

$$\frac{v^2-250}{150} = e^{-2} \quad (\text{take exponentials of both sides})$$

$$v^2 = 150e^{-2} + 250$$

$$= 270.30029...$$

$$v = 16.4408...$$

∴ The velocity on return to the ground is approximately 16.44 m/s

(b) Find the total time for the particle to return to the ground.

$$\ddot{x} = -0.04(v^2 - 250) \quad \text{from (i)}$$

$$\text{i.e. } \frac{dv}{dt} = -0.04(v^2 - 250)$$

$$\int_{20}^{16.44} \frac{dv}{v^2 - 250} = \int_0^T -0.04 dt$$

$$\text{NB } \frac{1}{v^2 - 250} = \frac{A}{v - \sqrt{250}} + \frac{B}{v + \sqrt{250}}$$

$$1 = A(v + \sqrt{250}) + B(v - \sqrt{250})$$

$$1 = v(A + B) + \sqrt{250}(A - B)$$

$$A - B = \frac{1}{\sqrt{250}} \quad A + B = 0$$

$$A = -B$$

$$2A = \frac{1}{\sqrt{250}} \Rightarrow A = \frac{1}{2\sqrt{250}} \quad \text{and } B = -\frac{1}{2\sqrt{250}}$$

$$\text{Now, } \frac{1}{2\sqrt{250}} \int_{20}^{16.44} \left(\frac{1}{v - \sqrt{250}} - \frac{1}{v + \sqrt{250}} \right) dt = -0.04 \left[t \right]_0^T$$

$$\frac{1}{2\sqrt{250}} \left[\ln \left(\frac{v - \sqrt{250}}{v + \sqrt{250}} \right) \right]_{20}^{16.44} = -0.04 [T - 0]$$

$$T = -\frac{25}{\sqrt{250}} \ln \left(\frac{16.44 - \sqrt{250}}{16.44 + \sqrt{250}} \times \frac{20 + \sqrt{250}}{20 - \sqrt{250}} \right)$$

$$= -0.79056 \times \ln \left(\frac{0.628611... \times 35.81138...}{32.25138... \times 4.18861...} \right)$$

$$= 1.41662...$$

$$\text{Total Time} = 1.42 + 5$$

$$= 6.42 \text{ seconds (to 2 d.p.)}$$

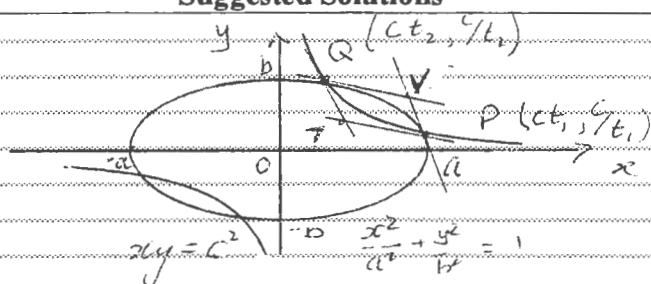
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Suggested Solutions

Marks

Marker's Comments

(i)


 (ii) The point $(ct, \frac{c}{t})$ lies on $xy = c^2$

and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{(ct)^2}{a^2} + \frac{(c/t)^2}{b^2} = 1$$

$$b^2 c^2 t^2 + a^2 c^2 = a^2 b^2 t^2$$

$$b^2 c^2 t^2 + a^2 c^2 - a^2 b^2 t^2 = 0$$

 (iii) Equation of tangent is $x + t^2 y = 2ct$

at P: $x + t_1^2 y = 2ct_1$ (i)

at Q: $x + t_2^2 y = 2ct_2$ (ii)

$$y(t_1^2 - t_2^2) = 2c(t_1 - t_2)$$

$$y(t_1 - t_2)(t_1 + t_2) = 2c(t_1 - t_2)$$

$$y = \frac{2c}{t_1 + t_2}$$

Sub into (i)

$$x = 2ct_1 - t_1^2 \left[\frac{2c}{t_1 + t_2} \right]$$

$$= 2c \frac{t_1^2 + t_1 t_2 - t_1^2}{t_1 + t_2}$$

$$x = \frac{2ct_1 t_2}{t_1 + t_2}$$

$$T = \left[\frac{2ct_1 t_2}{t_1 + t_2}, \frac{2c}{t_1 + t_2} \right]$$

 (iv) Equation of tangent is $bx + a^2 y_1 y = a^2 b^2$

at P: $bct_1 x + \frac{a^2 c}{t_1} y = a^2 b^2 \times t_1$

at Q: $bct_2 x + \frac{a^2 c}{t_2} y = a^2 b^2 \times t_2$

③ $bct_1 t_2 x + a^2 c t_2 y = a^2 b^2 t_2$

④ $bct_1 t_2 x + a^2 c t_1 y = a^2 b^2 t_1$

③ - ④ $a^2 y y \left[\frac{t_2}{t_1} - \frac{t_1}{t_2} \right] = a^2 b^2 [t_2 - t_1]$

$$a^2 c \left[\frac{t_2^2 - t_1^2}{t_1 t_2} \right] = a^2 b^2 [t_2 - t_1]$$

①

① ② for correct position of P and Q

① ② for V and T

②

 ① sub $(ct, \frac{c}{t})$ into $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

① simplifying

②

 no loss of mark if $t_1 \neq t_2$ not written

 $t_1 \neq t_2$

① x coordinate

① y coordinate

Suggested Solutions

Marks

Marker's Comments

$$y = \frac{b^2 t_1 t_2 (t_2 - t_1)}{c (t_1 - t_2) (t_1 + t_2)} \quad t_1 \neq t_2$$

$$= \frac{b^2 t_1 t_2}{c (t_1 + t_2)}$$

$$(i) \times t_1 \quad (5) \quad b^2 c t_1^2 x + a^2 c y = a^2 b^2 t_1$$

$$(ii) \times t_2 \quad (6) \quad b^2 c t_2^2 x + a^2 c y = a^2 b^2 t_2$$

$$(5) - (6) \quad b^2 c x [t_1^2 - t_2^2] = a^2 b^2 [t_1 - t_2]$$

$$x = \frac{a^2 (t_1 - t_2)}{c (t_1 - t_2) (t_1 + t_2)} = \frac{a^2}{c (t_1 + t_2)}$$

$$V = \left[\frac{a^2}{c (t_1 + t_2)}, \frac{b^2 t_1 t_2}{c (t_1 + t_2)} \right]$$

$$(v) \text{ Gradient of OT } m_{OT} = \frac{2c}{t_1 + t_2} / \frac{2c t_1 t_2}{t_1 + t_2} = \frac{1}{t_1 t_2}$$

$$\text{Gradient of OV } m_{OV} = \frac{b^2 t_1 t_2}{c (t_1 + t_2)} / \frac{a^2}{c (t_1 + t_2)} = \frac{b^2}{a^2} [t_1 t_2]$$

$$\text{Roots of } b^2 c^2 t^4 - a^2 b^2 t^2 + a^2 c^2 = 0$$

t_1, t_2 and $-t_1, -t_2$ by symmetry

$$\text{product of roots } t_1^2 t_2^2 = \frac{a^2 c^2}{b^2 c^2}$$

$$\therefore t_1 t_2 = a/b \text{ as } t_1 t_2 > 0$$

$$\therefore m_{OV} = \frac{b^2}{a^2} \times \frac{a}{b} = \frac{b}{a}$$

$$m_{OT} = \frac{1}{t_1 t_2} = \frac{b}{a}$$

$\therefore V, O, T$ collinear (2 equal gradients and common point)

$\therefore VT$ passes through origin

Alternatively Equation of TV

$$y - \frac{2c}{t_1 + t_2} = \frac{b^2 t_1 t_2 - 2c^2}{a^2 - 2c^2 t_1 t_2} \left[x - \frac{2c t_1 t_2}{t_1 + t_2} \right]$$

$$\text{LHS} = \text{RHS when } x = 0, y = 0 \text{ and } t_1 t_2 = a/b$$

$\therefore TV$ passes through origin

②

① x coordinate

① y coordinate

① gradients
OT, OV.

$$① \quad t_1 t_2 = a/b.$$

① Conclusion
with working
Alternatively

① Gradient TV

① Equation of TV
and sub (0,0)

① showing correctly
LHS = RHS

(using $t_1 t_2 = a/b$)

Suggested Solutions

Marks

Marker's Comments

(v1) Focus = $(c\sqrt{2}, c\sqrt{2})$

(2)

$$x = \frac{a^2}{c(t_1+t_2)} = c\sqrt{2}$$

$$y = \frac{b^2 t_1 t_2}{c(t_1+t_2)} = c\sqrt{2}$$

$$\therefore \frac{x}{y} = \frac{a^2}{b^2 t_1 t_2} = 1$$

$$a^2 = b^2 t_1 t_2 \quad t_1 t_2 = \frac{a^2}{b^2}$$

$$a^2 = b^2 \frac{a^2}{b^2} \quad a^2 = ab$$

$$\therefore a = 1 \quad a \neq 0, b \neq 0$$

$$a = b$$

Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \therefore \frac{x^2}{a^2} + \frac{y^2}{a^2} = 1$

$\therefore x^2 + y^2 = a^2$
circle centre (0,0) radius a units

(β) Focus lies on tangent to ellipse
 $b^2 x x_1 + a^2 y y_1 = a^2 b^2$ (at x_1, y_1)

$$a^2 x x_1 + a^2 y y_1 = a^4 \quad a=b$$

$$x x_1 + y y_1 = a^2$$

Tangent passes through focus $(c\sqrt{2}, c\sqrt{2})$

$$c\sqrt{2} x_1 + c\sqrt{2} y_1 = a^2$$

$$x_1 + y_1 = \frac{a^2}{c\sqrt{2}}$$

But $x_1^2 + y_1^2 = a^2$ (circle).

$$x_1^2 + 2x_1 y_1 + y_1^2 = a^2 + 2x_1 y_1$$

$$(x_1 + y_1)^2 = a^2 + 2c^2 y_1 \quad \text{as } x_1, y_1 \text{ is on } xy = c^2$$

$$\frac{a^4}{2c^2} = a^2 + 2c^2$$

$$a^4 = 2c^2 a^2 + 4c^4$$

$$a^4 - 2c^2 a^2 - 4c^4 = 0$$

$$a^2 = \frac{2c^2 \pm \sqrt{4c^4 + 16c^4}}{2}$$

$$a > 0 \quad a = \frac{2c^2 + c^2 \sqrt{20}}{2} = c^2 + c^2 \sqrt{5}$$

$$a > 0 \quad c > 0 \quad a = c \sqrt{1 + \sqrt{5}}$$

① relating a and b

① showing $a=b$
(with proof)

Other methods possible.

① expression for $x_1 + y_1$

① Quadratic Equation in a^2

① Solution

MATHEMATICS Extension 2 : Question 16

Suggested Solutions	Marks	Marker's Comments
16 a) $I_n = \int_0^{2\pi} (1 + \cos \theta)^n d\theta, n \geq 0$ (i) Show $I_{n+1} = \frac{2n+1}{n+1} I_n$ $I_{n+1} = \int_0^{2\pi} (1 + \cos \theta)^{n+1} d\theta$ $= \int_0^{2\pi} (1 + \cos \theta)(1 + \cos \theta)^n d\theta$ $= \int_0^{2\pi} (1 + \cos \theta)^n d\theta + \int_0^{2\pi} \cos \theta (1 + \cos \theta)^n d\theta$ Integrating by parts $u = (1 + \cos \theta)^n \quad v' = \cos \theta$ $u' = -n(1 + \cos \theta)^{n-1} \sin \theta \quad v = \sin \theta$ $I_{n+1} = I_n + \left[\int_0^{2\pi} (1 + \cos \theta)^n \sin \theta d\theta + n \int_0^{2\pi} \sin^2 \theta (1 + \cos \theta)^{n-1} d\theta \right]$ $= I_n + \left[(1 + \cos 2\pi)^n \sin 2\pi - (1 + \cos 0)^n \sin 0 \right]$ $+ n \int_0^{2\pi} (1 - \cos^2 \theta) (1 + \cos \theta)^{n-1} d\theta$ $= I_n + 0 - n \int_0^{2\pi} (\cos^2 \theta + 2 \cos \theta - 2 \cos \theta + 2 - 2 - 1) (1 + \cos \theta)^{n-1} d\theta$ $= I_n - n \int_0^{2\pi} [(1 + \cos \theta)^2 - 2(1 + \cos \theta)] (1 + \cos \theta)^{n-1} d\theta$ $= I_n - n \int_0^{2\pi} (1 + \cos \theta)^{n+1} - 2(1 + \cos \theta)^n d\theta$ $I_{n+1} = I_n - n I_{n+1} + 2n I_n$ $(n+1)I_{n+1} = (2n+1)I_n$ $I_{n+1} = \frac{2n+1}{n+1} I_n \quad \#$ (ii) Find I_3 : $I_0 = \int_0^{2\pi} (1 + \cos \theta)^0 d\theta = \int_0^{2\pi} d\theta = 2\pi$ $I_1 = I_0 = 2\pi$ $I_2 = \frac{2+1}{1+1} \cdot 2\pi = 3\pi$ $I_3 = \frac{2(2)+1}{2+1} \cdot 3\pi = 5\pi$ $\therefore I_3 = 5\pi \quad \#$		

MATHEMATICS Extension 2 : Question... 16

Suggested Solutions	Marks	Marker's Comments
16(b)(i)		(1)
(ii) $\angle PLA + \angle PMA = 90^\circ + 90^\circ$ (L & M are the feet of the perpendiculars from P to AB & AC respectively) $= 180^\circ$ $\therefore PMAL$ is a cyclic quadrilateral (opposite angles are supplementary) $\therefore P, M, A$ and L are concyclic points		(2)
(iii) $\angle PMC = \angle PNC = 90^\circ$ (M & N are the feet of the perpendiculars from P to AC & CB resp'y) $\therefore PCNM$ is a cyclic quadrilateral (angles subtended by interval PC on the same side are equal) $\therefore P, C, N$ & M are concyclic points		(2)
(iv). Show: L, M and N are collinear		
Constructions: Join ML, MN, PA & PC.		
Proof: $\angle PCB = \angle PAL$ (exterior angle of cyclic quad. $PABC$ equals the interior opposite angle) $\angle PAL = \angle PML$ (angles at the circumference in the same segment of cyclic quad. $PMAL$) are equal		(1)
$\therefore \angle PCB = \angle PML$		(1)
Also, $\angle PCN = \angle PMN$ (exterior angle of cyclic quad. $PCMN$ equals the interior opposite angle)		(1)
$\angle PML + \angle PMN = \angle PCB + \angle PCN$ $= 180^\circ$ (straight angle BCD equals 180°)		
Now $\angle PML + \angle PMN = \angle LMN = 180^\circ$ $\therefore LMN$ is a straight angle		
$\therefore L, M$ & N are collinear		(1)